QUANTITATIVE RESEARCH SPECIFICATION

Multi-Timeframe ML Price Action Trading System

Dynamic Level Pruning, Feature Invariance, Tunable Ensembles & Data Ingestion Protocol

Architecture Intent: This document serves as the operational handbook for institutional cryptocurrency modeling across 15m, 1h, 4h, 1d, and 1w intervals. It resolves lower-timeframe saturation, enforces candle-count invariant windows, dynamic hyperparameter tuning for ensemble weighting, and strict leak-free time-series cross-validation.

1. The Problem Space: Structural Pitfalls in ML Price Action

Traditional price action indicators and retail automated tools fail in machine learning production due to three mathematical traps:

- **Level Over-Saturation & Historical Decay:** Retaining every 15m or 1h swing high/low over multi-month regimes creates a continuous dense grid of lines where nearly every candle intersects a residual level, extinguishing predictive signal.
- **Non-Stationarity of Raw Levels:** Absolute price coordinates (e.g., $65,400 vs. $19,800) cannot be evaluated by tree-based or sequential models. Features must be transformed into percentage-normalized distances and state flags.
- **Microstructure Noise vs. Sample Size Dilemma:** Sub-hourly bars carry exchange spread slippage and execution anomalies, whereas macro bars (1D/1W) yield too few trades, inducing extreme model overfitting.

2. Timeframe Hierarchy & Dynamic Bar-Count Bounds

Calendar days (e.g., "14 days") distort logic across volatility expansions and consolidations. All model memory is standardized strictly by candle counts:

TIMEFRAME	MODEL DESIGNATION	BAR LOOKBACK	MAX ACTIVE CAP	PRUNING & INVALIDATION
Weekly (1W)	Macro Cycle Regime Gate	100 – 150 bars	3 – 5 levels	Structural break past 2× ATR
Daily (1D)	Swing Trend & Filter	180 – 360 bars	5 – 7 levels	Full bar close through zone
4-Hour (4H)	Setup Anchor & S/D Pools	180 – 250 bars	6 – 8 levels	Zone mitigated or > 4× ATR away
1-Hour (1H)	Intermediate Liquidity	100 – 150 bars	4 – 6 levels	Zone mitigated or > 3× ATR away
15-Min (15m)	Micro Entry Trigger	96 – 192 bars	2 – 4 levels	**Single-Use:** Purged on 1st touch

3. Dynamic Tunable Ensemble Matrix (Meta-Model Layer)

Hardcoded weights (e.g., fixed 25% or 30% rules) fail across shifting regimes. The ensemble composite score is dynamically tuned via cross-validation hyperparameter search:

Dynamic Ensemble Objective Function:

$S_{composite} = w_{1W} \cdot S_{1W} + w_{1D} \cdot S_{1D} + w_{4H} \cdot S_{4H} + w_{1H} \cdot S_{1H} + w_{15m} \cdot S_{15m}$

Subject to: $\sum w_i = 1.0$, with weights bounded by $w_i \geq 0$. Weights and threshold triggers (θ_{buy}, θ_{sell}) are continuously tuned using Bayesian Optimization (Optuna) or Logistic Meta-Regression over rolling historical validation windows.

While the weights adapt dynamically to maximize out-of-sample Sharpe ratio, the trading engine maintains three deterministic risk invariants that machine learning cannot override:

- **Macro Permissive Gating:** If the combined higher timeframe directional score is negative, the engine unconditionally suppresses long triggers regardless of short-term 15m confluence.
- **Zone Trigger Precondition:** Micro 15m or 1h entry patterns are only evaluated when market price has tapped an active, unmitigated 4H or 1D institutional level.
- **Structural Stop-Loss Enforcement:** Invalidation coordinates are mapped strictly to physical price pivots rather than model predictions.

4. Mathematical Feature Invariance & Stationarity

Raw dollar quotes are converted to invariant, stationary geometric metrics before entering any training pipeline:

1 Relative Support Distance

$d_{sup} = (Close - Level_{sup}) / Close$

Normalized percentage distance to nearest active higher-timeframe support zone.

2 Relative Resistance Distance

$d_{res} = (Level_{res} - Close) / Close$

Normalized percentage distance to nearest active higher-timeframe resistance zone.

3 Multi-Timeframe Confluence

$C_{idx} = \sum 1(\text{Overlap within } 0.35\%)$

Integer count measuring how many timeframes share overlapping levels within a narrow threshold.

4 Structural Zone Age & Touch

$Age = N_{candles}$ and $T_{count} = \sum Tests$

Tracks candle lifespan and touch counts; levels degrade in probability as touches increase.

5. Authoritative Data Sourcing & Archive Architecture

Model accuracy relies strictly on raw tick data from deep institutional matching engines. Third-party aggregators introduce interpolation artifacts and missing candle intervals:

SOURCE PROVIDER	RELIABILITY	ACCESS COST	OPTIMAL SYSTEM APPLICATION
data.binance.vision	10 / 10	Free Public	Primary multi-year backfill (USDT-M perpetual klines). Zero interpolation gaps.
ccxt (Exchange API)	9 / 10	Free Public	Live operational ingestion, automated bar closures, and forward inference polling.
Coinglass / CryptoQuant	9 / 10	Freemium	Derivatives context features: Open Interest delta, funding velocity, liquidations.
Tardis.dev	10 / 10	Commercial	Microstructure backfilling: tick-level L2/L3 order book snapshots and delta depth.

6. Production Training & Validation Protocol

Financial time-series data requires non-standard cross-validation to prevent forward-looking leakage and alpha decay:

- **Purged & Embargoed Walk-Forward Splits:** Never utilize standard random K-Fold splits. Apply a rolling 24-month training window followed by a 1-week purge buffer and a 3-month out-of-sample forward test fold.
- **Triple Barrier Labeling:** Rather than predicting next-bar direction, labels are generated dynamically via volatility bounds:
  • Upper Profit Take: $+2.0 \times ATR_{4H}$
  • Lower Stop Invalidation: $-1.0 \times ATR_{4H}$
  • Vertical Time Barrier: Exit position after 6 bars (24 hours).
- **Hyperparameter Search Protocol:** Optimize tree depth, learning rate, and ensemble weights simultaneously over validation folds using cross-validated information ratio rather than raw directional accuracy.

> **Execution Rule of Thumb:**
> Gradient-boosted decision trees (XGBoost, LightGBM) consistently outperform deep sequential neural networks (LSTM, Transformers) on tabular 4H price action due to superior noise tolerance and robust split-point handling on sparse feature spaces.